CHE 32000 Statistical Modeling and Quality Enhancement Spring 2026

Class Times: Lectures Mon, Wed (and occasional Fri): 11:30-12:20 FRNY G140
Recitation/Computer Lab: As scheduled

Instructors: Kendall Thomson (Instructor)
FRNY 1052
Office Hours: Wed, 12:45-1:45
thomsonk@purdue.edu

Kurt Ristroph (Instructor)
ABE 2020
Office Hours: TBA
ristroph@purdue.edu

A. Graduate Teaching Assistants:

Luis Guzman Julio, lguzmanj@purdue.edu
Shraman Pal, pal52@purdue.edu
Hsuan-Han Chiu, chiu137@purdue.edu
Haripriyan Uthayakumar, huthayak@purdue.edu

The TAs for your computer lab recitation will give you further contact details and office hours in your first lab class.

B. Course Objective
Develop and apply statistical methods, including designing experiments and building models from experimental data for use in the senior chemical engineering laboratories and for professional use in the chemical process industries.

C. Course Outcomes (numbers in parentheses refer to related program educational objective)

1. Understand and use the concepts of probability for use with the normal and t-distributions, confidence intervals, and hypothesis testing (1, 2).
2. Understand and apply least squares fitting of data by use of linear algebra with computers (1, 2, 5).
3. Understand and apply regression and model building from data with experimental error obtained with replicates to establish an adequate model (1, 2, 5).
4. Design experiments for single and multiple factors and apply regression to model the results (1, 2, 5).
5. Understand and apply propagation of error and sources of variation or variance components (1, 2).
6. Understand and apply statistical quality control and improvement with various charts such as average, range, and cusum (concepts of consumer and producer risk and capability index are included) (1, 2, 5, 8).
7. Work professionally and ethically in teams (6, 7).

D. Text: *Engineering Statistics 5th,* D. C. Montgomery, G. C. Runger, and N. R. Hubele, John Wiley & Sons, Inc., 2011.

E. Software: During this class, students will gain a working knowledge of *JMP* and make some use of *Excel*.

F. Grading Policy:

Homework Assignments	100 pts (10 each)
Computer Project	150 pts
Three 2-hour Exams	300 pts (100 each)
Participation	50 pts
Course Survey	10 pts
Total	610 pts

G. Course Grading

Final grades for this class will be assigned using the +/- system (A+, A, A-, B+, B, etc…)

H. Homework:

Assignments will be available from Brightspace most Friday mornings and are due in completed form by the following Friday at 11:59 PM (submitted online in pdf form). **Late homework will not be accepted**. While you may find it helpful to discuss problem sets with one another, *what you turn in must be your own work.* **Your homeworks are to be neat and legible. Write on one side of the paper.**

I. Exam Make-up Policy

Occasionally students will have to miss an intra-semester exam for personal or uncontrollable reasons. However, if an exam needs to me missed, **there will NOT be a make-up exam given during the regular semester**. Instead, the final exam will act as a make-up exam.

PLEASE NOTE: A student can only miss a midterm exam for a legitimate reason (death in the family, illness, emergency, etc.) and **ONLY** if given permission by Professor Thomson. If a student attends all three exams during the semester, the final exam will not have to be taken.

PLEASE NOTE: You **cannot** take the final exam in order to replace a poor grade on a mid-term exam. The final exam is ONLY for students who had a legitimate and excused absence during the semester.

J. On-line Course Evaluation

It is important for department and instructors to receive thorough feedback on all courses taught, so it is your responsibility to provide such feedback. Participation in the on-line course evaluation is mandatory and will be treated as a homework assignment worth 15 pts.

K. Computer Project:

There will be one computer project in the second half of the semester. The computer project will be done in small teams. Further details of the project will be supplied after the mid semester break.

L. Student Expectations.

This is a 3 credit hour course, and it is expected that each student will spend 12 hours each week working on homework, studying, and reading the course text (3 hours/credit). This class, and all subsequent chemical engineering classes, will be much more challenging than you are likely used to, and at the same time much more rewarding than anything that most students will have seen before. We encourage you to take studying seriously and establish good study habits such as previewing the reading material ***before*** the lectures and practicing additional problems.

M. Instructors' Commitment.

Your instructors will: 1) be courteous, punctual, well-organized, and prepared for lecture and other class activities; 2) answer questions clearly in class or arrange for detailed discussions out of class if in-class answers are not suitably clear; 3) be available during office hours or notify you beforehand if we are unable to keep them; 4) provide a suitable guest lecturer when we are traveling; and 5) grade uniformly and consistently to the posted guidelines.

Email: Given that Purdue is a public institution, with published emails for all employees, I get an overwhelming amount of spam. If you need to get in contact with me via email, please include "CHE 320" in the subject line, so an appropriate filter will pick it up. I will try to answer all emails as soon as I can, however, please note I will not respond to email questions about items that I have repeatedly discussed in class and have posted as announcements on Brightspace. It is your responsibility to show and listen during lecture and to check Brightspace frequently.

N. Consulting with the Faculty Members.

We encourage you to discuss academic or personal questions with us during our office hours or via email. These discussions need not be limited to ChE 32000 content.

O. Academic Dishonesty.

Academic dishonesty ***will not be tolerated*** in any form in this course. Specifically, Purdue prohibits "dishonesty in connection with any University activity. Cheating, plagiarism, or knowingly furnishing false information to the University are examples of dishonesty." [Section B-2-a, Code of Student Conduct] Furthermore, the University Senate has stipulated that "the commitment of acts of cheating, lying, and deceit in any of their diverse forms (such as the use of substitutes for taking examinations, the use of illegal cribs, plagiarism, and copying during examinations) is dishonest and must not be tolerated. Moreover, knowingly to aid and abet, directly or indirectly, other parties in committing dishonest acts is in itself dishonest." [University Senate Document 72-18, December 15, 1972] All incidents of academic dishonesty will be reported to the Dean of Students. **Such incidents include: i) possessing or accessing, in hardcopy or electronic form, the solution manual to the course text, or to the exams, ii) claiming credit for work that is not your own original work, and iii) enabling other students to create work that is not their original work. The punishment for a first offense on a homework assignment is a grade of zero for the entire work and the punishment for a second offense is an F grade for the course. The punishment for any offense for an exam is an F grade for the course.**

P. Conduct.

University policy states that it is the responsibility of all students to attend all class sessions (http://www.purdue.edu/studentregulations/regulations_procedures/classes.html). Each student is expected to come to class on time and not disrupt the class. Each student is also expected to follow Purdue's codes of student conduct (http://www.purdue.edu/studentregulations/student_conduct/regulations.html) and behave in a professional manner. The rights of students in violation of the code of conduct are outlined. Each student is expected to exhibit consideration and respect towards the other students, the graders, the teaching assistants (TAs), and the faculty member. Each student is expected to exhibit a positive attitude. Your conduct will be a factor in awarding grades to students between two letter grades. Purdue University's student conduct policy specifically addresses academic dishonesty.

Q. Violent Behavior Policy.
Purdue University is committed to providing a safe and secure campus environment for members of the University community. Purdue strives to create an educational environment for students and a work environment for employees that promote educational and career goals. Violent behavior impedes such goals. Therefore, violent behavior is prohibited in or on any University Facility or while participating in any University activity.

R. Nondiscrimination.
Purdue University is committed to maintaining a community which recognizes and values the inherent worth and dignity of every person; fosters tolerance, sensitivity, understanding, and mutual respect among its members; and encourages each individual to strive to reach his or her own potential. In pursuit of its goal of academic excellence, the University seeks to develop and nurture diversity. The University believes that diversity among its many members strengthens the institution, stimulates creativity, promotes the exchange of ideas, and enriches campus life. Purdue University prohibits discrimination against any member of the University community on the basis of race, religion, color, sex, age, national origin or ancestry, genetic information, marital status, parental status, sexual orientation, gender identity and expression, disability, or status as a veteran. The University will conduct its programs, services and activities consistent with applicable federal, state and local laws, regulations and orders and in conformance with the procedures and limitations as set forth in Executive Memorandum No. D-1, which provides specific contractual rights and remedies. Any student who believes they have been discriminated against may visit www.purdue.edu/report-hate to submit a complaint to the Office of Institutional Equity. Information may be reported anonymously.

S. Emergency Preparedness. Purdue University is a very safe campus and there is a low probability that a serious incident will occur here at Purdue. However, it is important to emphasize the emergency procedures for evacuation and shelter-in-place incidents. Preparedness will be critical if an unexpected event is to occur. Emergency preparedness is your personal responsibility. Purdue University is actively preparing for natural disasters or human-caused incidents with the ultimate goal of maintaining a safe and secure campus. The following is a review of the emergency procedures at Purdue University.

1. For any emergency call 911.
2. There are nearly 300 Emergency Telephone Systems throughout campus that connect directly to the Purdue Police Department (PUPD). If you feel threatened or need help, push the button and you will be connected to the PUPD.
3. If there is a fire alarm, we will immediately evacuate the building and proceed to in front of the MSEE building. Do not use the elevator.
4. If there is a Shelter-in-Place requirement for a tornado warning, we will shelter in the lowest level of this building away from windows and doors. This location is between FRNY G140 and FRNY B124.
5. If there is a Shelter-in-Place requirement for a hazardous materials release, we will shelter in the classroom shutting any open doors and windows.
6. If there is a Shelter-in-Place requirement for a civil disturbance, we will shelter in a room that is securable preferably without windows. This location is FRNY 2182.

T. Course Meeting Schedule.

Lectures:	M, W, (some Fridays)	11:30a – 12:20p	FRNY G140
Recitations:	see schedule	see schedule	
Examination 1:	2/18 (Wednesday)	11:30a – 12:20p	FRNY G140
Examination 2:	4/1 (Wednesday)	11:30a – 12:20p	FRNY G140
Examination 3:	4/22 (Wednesday)	11:30a – 12:20p	FRNY G140

If you have a scheduled class or other reasonable issue (*e.g.*, religious observance) that conflicts with a scheduled exam, please tell Professor Thomson as soon as possible so that plans may be made for you to take the exam at an alternate time. It is not acceptable for you to skip a scheduled class meeting in order to take a ChE 32000 exam.

No Lecture or Rec:	1/19 (Mon)	MLK Jr. Holiday
No Lecture or Rec:	3/16-3/20	Spring Break

U. Attendance. University policy states that it is the responsibility of all students to attend all class sessions (including online content):
(http://www.purdue.edu/studentregulations/regulations_procedures/classes.html)
You are expected to attend all lectures and recitation periods.

Only the instructors can excuse a student from a course requirement or responsibility. When conflicts can be anticipated, such as for many University-sponsored activities and religious observations, the student should inform the instructor of the situation as far in advance as possible. For unanticipated or emergency conflict, when advance notification to an instructor is not possible, the student should contact the instructor as soon as possible by email, through Brightspace, or by phone. When the student is unable to make direct contact with the instructor and is unable to leave word with the instructor's department because of circumstances beyond the student's control, and in cases of bereavement, quarantine, or isolation, the student or the student's representative should contact the Office of the Dean of Students via email or phone at 765-494-1747. Our course Brightspace includes a link on Attendance and Grief Absence policies under the University Policies menu.

V. Illness. If you become quarantined or isolated at any point in time during the semester, in addition to support from the Protect Purdue Health Center, you will also have access to an Academic Case Manager who can provide you academic support during this time. Your Academic Case Manager can be reached at acmq@purdue.edu and will provide you with general guidelines/resources around communicating with your instructors, be available for academic support, and offer suggestions for how to be successful when learning remotely. Importantly, if you find yourself too sick to progress in the course, notify your academic case manager and notify me via email or Brightspace. We will make arrangements based on your particular situation. The Office of the Dean of Students (odos@purdue.edu) is also available to support you should this situation occur.

W. Bereavement Policy. Purdue recognizes that a time of bereavement is very difficult for a student. The University therefore provides rights to students facing the loss of a family member through the Grief Absence Policy for Students (GAPS):
http://www.purdue.edu/odos/services/griefabsencepolicyforstudents.php.
Students who find themselves in need of assistance in a time of bereavement should contact Professor Thomson privately to discuss specific needs.

X. Campus Emergencies. In the event of a major campus emergency, course requirements, deadlines and grading percentages are subject to changes that may be necessitated by a revised semester calendar or other circumstances beyond the instructors' control. ***Here are ways to get information about changes in this course. You are expected to check your @purdue.edu email address frequently.***

Y. Individual Learning and Testing Needs. Purdue University strives to make learning experiences as accessible as possible. If you anticipate or experience physical or academic barriers based on disability, you are welcome to let me know so that we can discuss options. You are also encouraged to contact the Disability Resource Center at drc@purdue.edu or by phone: 765-494-1247.

In addition to the University policy, the Davidson School of Chemical Engineering has established procedures for students seeking accommodations. These can be found online at the ChE Undergrad Office website. Only those accommodation requests that conform to both University and ChE policy guidelines will be implemented.

DRC Exam Accommodations: All accommodations for exams must be scheduled on the actual exam date, during the actual exam time (or as close as possible). All requests for exam accommodations must be made at least **one week** before the scheduled date of the exam. We will deny any accommodation that does not follow these regulations.

This syllabus is subject to change without notice based on circumstances that arise which might have impacts on student/staff health or campus emergency procedures and/or distance learning capabilities.

ABET Syllabus

1. **CHE 32000 – Statistical Modeling**

2. **Credits and contact hours:**
   3 credits
   Lecture – 3 days per week at 50 minutes for 15 weeks.

3. **Instructor's or course coordinator's name: Kendall Thomson**

4. **Textbook(s):** *Engineering Statistics 5th,* D. C. Montgomery, G. C. Runger, and N. R. Hubele, John Wiley & Sons, Inc., 2011.
   a. Other supplemental materials: none

5. **Specific course information**
   a. Catalog description: Statistical modeling methods, design of experiments, error analysis, curve fitting and regression, analysis of variance, confidence intervals, quality control and enhancement: emphasizes preparation for designing chemical engineering laboratory experiments and analyzing data.
   b. Prerequisites: CHE 20500, Math Selective I (concurrent)
   c. Course status: Required

6. **Specific goals for the course**
   a. Specific outcomes of instruction:
   - Utilize the concepts of probability for use with the normal and t distributions, confidence intervals, and hypothesis testing (→43500)
   - Apply least squares fitting of data by use of linear algebra with computers (20500→); (→34800, 37700, 37800, 43500)
   - Apply regression and model building from data with experimental error obtained with replicates to establish an adequate model (→34800, 37700, 37800, 43500)
   - Design experiments for single and multiple factors and apply regression to model the results (→43500)
   - Apply propagation of error and sources of variation or variance components (→34800, 37700, 37800, 43500)
   - Apply statistical quality control and improvement with various charts such as average, range, and cusum (concepts of consumer and producer risk and capability index are included) (→43500)

   b. Student outcomes addressed by the course:
   1) Apply principles of engineering, science, and mathematics to solve complex chemical engineering problems.
   2) Apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.
   6) Develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.
   7) Acquire and apply new knowledge as needed, using appropriate learning strategies.

7. **Topics**
   statistical modeling
   probability theory
   discrete random variables & probability distributions
   continuous random variables & probability distributions
   joint probability distributions
   statistical inference
   central limit theorem

t-distribution/confidence limits
hypothesis testing
simple linear regression
nonlinear regression, correlation
multilinear regression
stepwise regression
experimental design
factorial experimentation
fractional factorial experimentation
response surface designs
process optimization
propagation of errors
statistical quality control

	Date	Chap. (required)	Lecture	
Week 1	1/12	1, 2, 3	Introduction	Kendall Thomson
	1/14	3	Probability/Discrete Distributions	Kendall Thomson
	1/16	3	Discrete Distributions	Kendall Thomson
Week 2	1/19		*Martin Luther King Jr. Holiday*	
	1/21	3	Discrete Distributions	Kendall Thomson
	1/23	3	Continuous Distributions	Kendall Thomson
Week 3	1/26	3	Continuous Distributions	Kendall Thomson
	1/28	4	Prop. of Error/Estimators	Kendall Thomson
	1/30	4	Estimators/Inferences on Mean I	Kendall Thomson
Week 4	2/2	4	Inferences on Mean I	Kendall Thomson
	2/4	4	Inferences on Mean II	Kendall Thomson
	2/6		*No class*	
Week 5	2/9	4	Inferences on Variance	Kendall Thomson
	2/11	5	Comparison of Two Means	Kurt Ristroph
	2/13	5	Comparison of Two Means	Kurt Ristroph
Week 6	2/16	5	Comparison of Variances	Kurt Ristroph
	2/18		**EXAM 1 (2/18)**	
	2/20		*No class*	
Week 7	2/23		*No class*	
	2/24	5	Comparison of Variances	Kurt Ristroph
	2/27	5	Analysis of Variance	Kurt Ristroph
Week 8	3/2	5	Analysis of Variance	Kurt Ristroph
	3/4	6	Simple Regression	Kurt Ristroph
	3/6		*No class*	
Week 9	3/9	6	Simple Regression	Kurt Ristroph
	3/11	6	Simple/Multiple Regression	Kurt Ristroph
	3/13	6	Multiple Regression	Kurt Ristroph
Week 10	3/16		*Spring Break*	
	3/18		*Spring Break*	
	3/20		*Spring Break*	
Week 11	3/23	7	Design of Experiments	Kendall Thomson
	3/25	7	Design of Experiments	Kendall Thomson
	3/27	7	Design of Experiments	Kendall Thomson
Week 12	3/30		Review	Kendall Thomson
	4/1		**EXAM 2 (4/1)**	
	4/3		*No class*	
Week 13	4/5	7	Design of Experiments/SRM	Kendall Thomson
	4/8	7	Surface Response Methods	Kendall Thomson
	4/10		*No class*	
Week 14	4/13	7	Project Introduction	Kendall Thomson
	4/15	7	Design of Experiments	Kendall Thomson
	4/17	7	Surface Response Methods	Kendall Thomson
Week 15	4/20	7	Surface Response Methods	
	4/22		**EXAM 3 (4/22)**	Kendall Thomson
	4/24		*No class*	
Week 16	4/27		Project Consulting	Kendall Thomson
	4/29		Project Consulting	Kendall Thomson
	5/1		**Projects Due 5/1**	Kendall Thomson

Homework and Computer Lab

Week of (Mon)	Homework Due (Fri)	Lab Participate
1/12	--	--
1/19	HW #1	Lab #1/2
1/26	HW #2	Lab #3/4
2/2	HW #3	Lab #5/6
2/9	HW #4	--
2/16	--	Lab #7
2/23	HW #5	Lab #8
3/2	--	--
3/9	HW #6	Lab #9
3/16	--	--
3/23	HW #7	Lab #10
3/30	HW #8	Lab #11
4/6	HW #9	Lab #12
4/13	--	Lab #13
4/20	HW #10	Lab #14
4/27	Project Due 5/1	